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Presentation of Final Project

Problem Statement

Machine Language Problem: *To develop a machine learning model that forecasts the Central Bank Rate in Uganda using historical interest rate data*.

Source: *The dataset was obtained from the Bank of Uganda and consists of daily interbank interest rate observations. The interbank rate reflects the cost at which commercial banks lend to each other in the short term and serves as an important indicator of liquidity conditions in the financial system. It can be accessed here: https://bou.or.ug/bouwebsite/bouwebsitecontent/statistics/InterestRates/Interest_rates.xlsx*

No. of rows and columns: Rows- 3,553 and Columns - 4

Data Description

Feature	Data Type	Description	Example
Date	Date	Date the interest rate was captured.	07/05/2012
Overnight (%)	Numeric	The weighted average interest rate charged by banks for unsecured overnight lending	20.85%
7-day (%)	Numeric	The average interest rate charged on short term maturity period of 7 days	22.52%
Overall(%)	Numeric	The average interbank lending rate across different short-term	22.38%

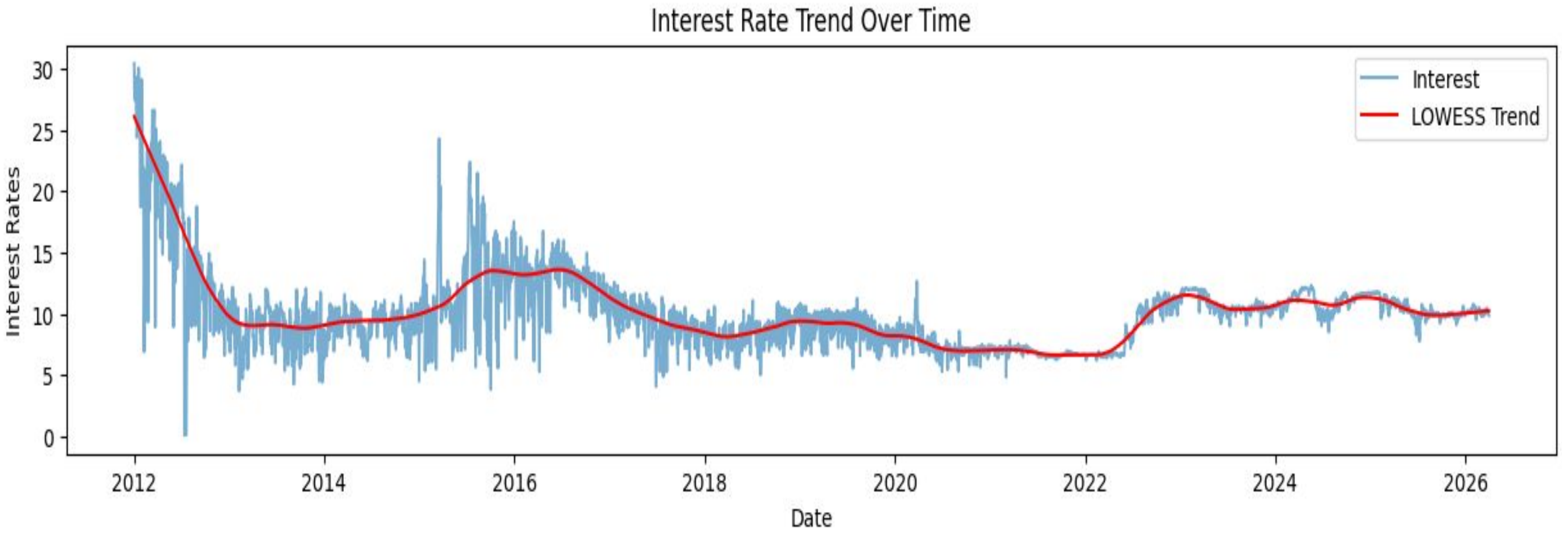
Data Exploration

The data was obtained from Bank of Uganda Website.

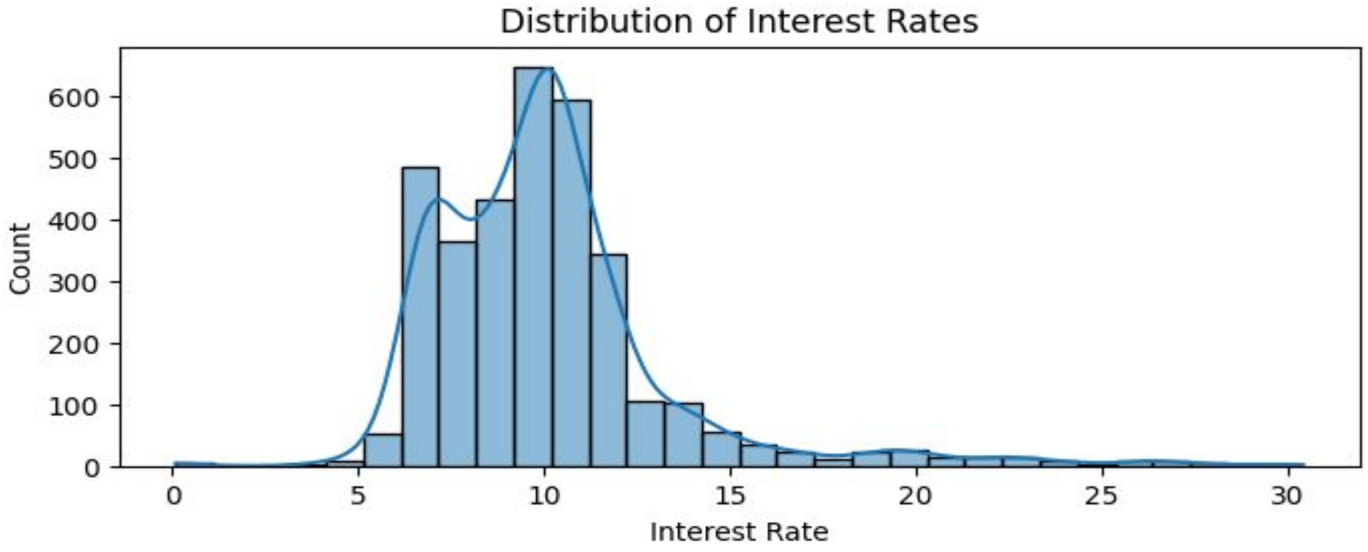
Time Frame: Jan 2012- March 2026

Missing Values: 0.0% for core %s.

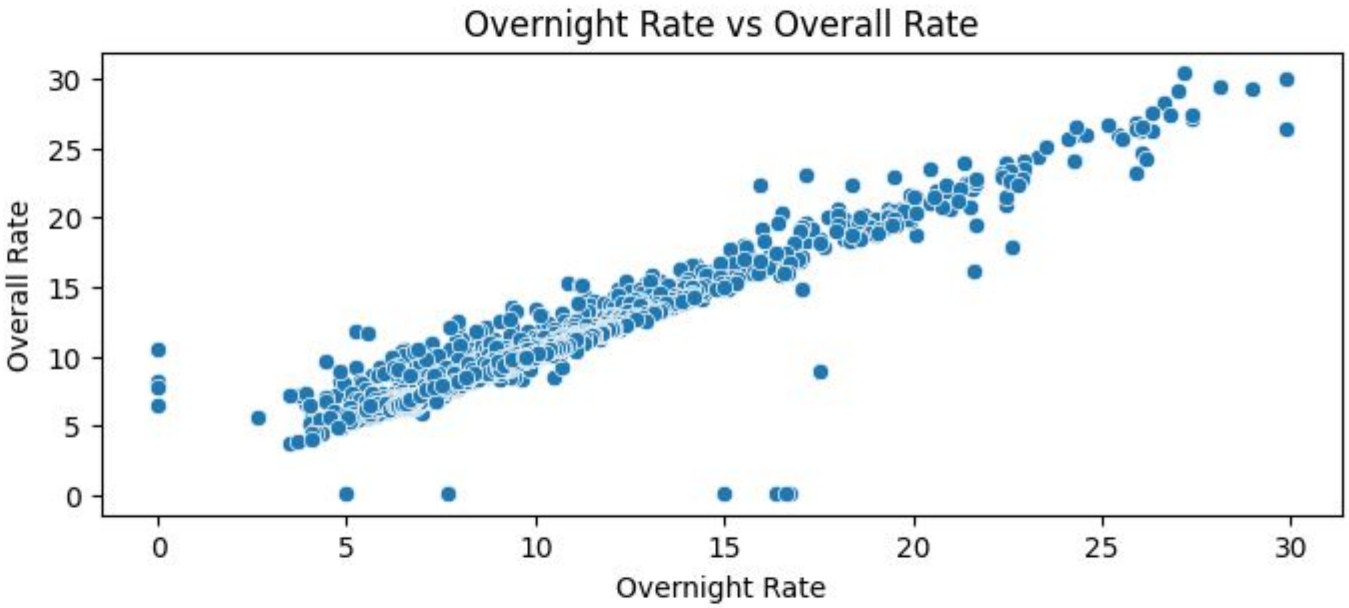
Interest Rate Trend over time



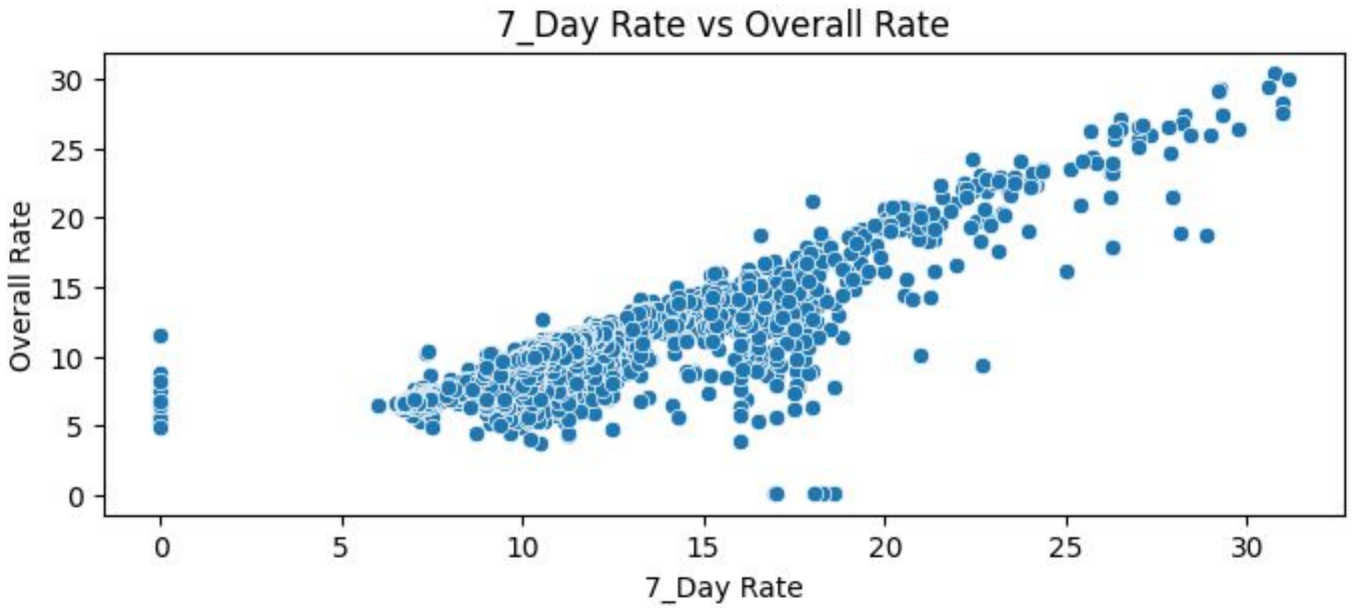
Distribution of Interest Rates



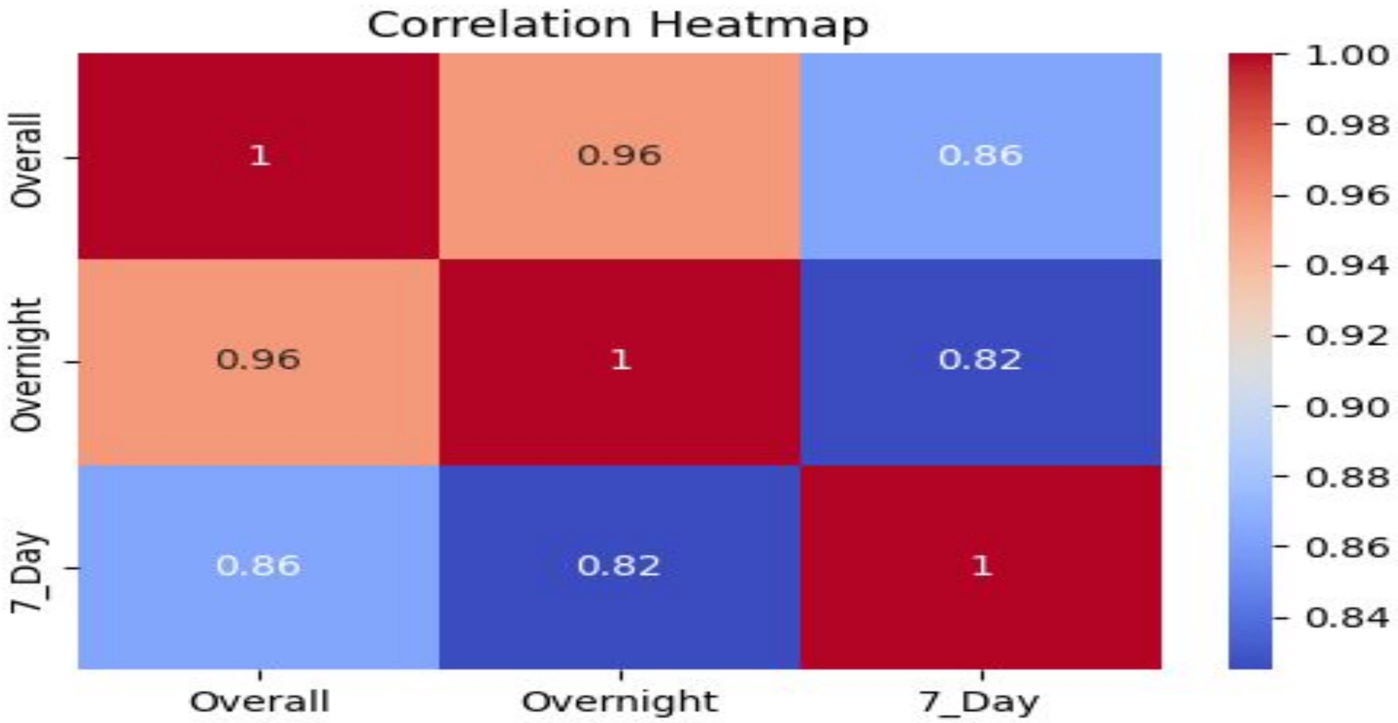
Overnight Rate Vs Overall Rate



7 Day Rate Vs Overall Rate



Correlation Heat Map



Key Observations from the data

1. There is a positive correlation between the 7-Day Rate, Overnight Rate and the Overall Rate
2. Through 2012–2014, the Bank of Uganda steadily reduced rates as inflation eased and macroeconomic conditions improved .
3. Between 2014 and 2019, Uganda experienced relatively moderate interest rates, mostly ranging between 9% and 17%.
4. The interest rate dropped in 2021 probably due to the effects of the COVID-19 pandemic on the economy.
5. Between 2022-2023, the interest rates rose from the pandemic lows.
6. Uganda has maintained steady interest rates for 2024-March 2026.